# THE EXTENDED ALTERNATING CECH COMPLEX REPRESENTS THE LOCAL COHOMOLOGY WITH SUPPORT PUSHED FORWARD FOR QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- A complex with cohomology sheaves supported on the closed subset Z is fixed by the functor i_* R H_Z, by the cohomology complexes-with-support lemma. invoked at proof lines 1200-1202. Identifies the alternating complex with its own local cohomology once supported on Z.
- The localization maps of the Cech complex are affine, so their direct images agree with derived direct images on quasi-coherent modules, by the relative affine vanishing lemma. invoked at proof lines 1209-1211 of the affine identification. Shows each term F_{i_0...i_p} is the derived pushforward of its restriction.
- Local cohomology with support in Z vanishes on modules supported in the open complement of Z, by the sections-support lemma. invoked at proof lines 1215-1217. Kills the higher Cech terms under R H_Z.

*Source reference.* Original remark 0G7I: sources/stacks-chow/tags.txt, source lines 20233–20233. Cited at statement lines 1186-1187 as the standing setup; the representation argument is inside the counted span.

*Source reference.* Original equation 0G7J: sources/stacks-chow/tags.txt, source lines 20234–20234. Cited at statement lines 1188-1189 as the complex asserted to represent the local cohomology; the identification is inside the counted span.

*Source reference.* Original remark 0G7L: sources/stacks-chow/tags.txt, source lines 20236–20236. Cited at proof lines 1197-1199 as the map to be shown an isomorphism; the identification argument is inside the counted span.

*Source reference.* Original lemma 0AEF: sources/stacks-chow/tags.txt, source lines 13123–13123. Cited at proof lines 1201-1202 for i_* R H_Z(F^bullet) = F^bullet; the assembly is inside the counted span.

*Source reference.* Original lemma 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cited at proof lines 1209-1210 of the term identification; the term computation is inside the counted span.

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*Date*: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0G73: sources/stacks-chow/tags.txt, source lines 20218–20218. Cited at proof lines 1216-1217 to kill the positive Cech terms; the assembly is inside the counted span.

**Lemma 0.2.** *With $X$, $f_1, \ldots, f_c \in \Gamma(X, \mathcal{O}_X)$, and $\mathcal{F}$ as in Remark 0G7I. If $\mathcal{F}$ is quasi-coherent, then the complex (0G7J) represents $i_*R\mathcal{H}_Z(\mathcal{F})$ in $D_Z(\mathcal{O}_X)$.*

*Proof.* Let us denote $\mathcal{F}^\bullet$ the complex (0G7J). The statement of the lemma means that the map $\mathcal{F}^\bullet \to i_*R\mathcal{H}_Z(\mathcal{F})$ of Remark 0G7L is an isomorphism. Since $\mathcal{F}^\bullet$ is in $D_Z(\mathcal{O}_X)$ (see remark cited), we see that $i_*R\mathcal{H}_Z(\mathcal{F}^\bullet) = \mathcal{F}^\bullet$ by Cohomology, Lemma 0AEF. The morphism $U_{i_0 \ldots i_p} \to X$ is affine as it is given over affine opens of $X$ by inverting the function $f_{i_0} \ldots f_{i_p}$. Thus we see that

$$\mathcal{F}_{i_0 \ldots i_p} = (U_{i_0 \ldots i_p} \to X)_*\mathcal{F}|_{U_{i_0 \ldots i_p}} = R(U_{i_0 \ldots i_p} \to X)_*\mathcal{F}|_{U_{i_0 \ldots i_p}}$$

by Cohomology of Schemes, Lemma 01XC and the assumption that $\mathcal{F}$ is quasi-coherent. We conclude that $R\mathcal{H}_Z(\mathcal{F}_{i_0 \ldots i_p}) = 0$ by Cohomology, Lemma 0G73. Thus $i_*R\mathcal{H}_Z(\mathcal{F}^p) = 0$ for $p > 0$. Putting everything together we obtain

$$\mathcal{F}^\bullet = i_*R\mathcal{H}_Z(\mathcal{F}^\bullet) = i_*R\mathcal{H}_Z(\mathcal{F})$$

as desired. □